MONASH University
Information Technology

Acknowledgements

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MONASH University
Information Technology

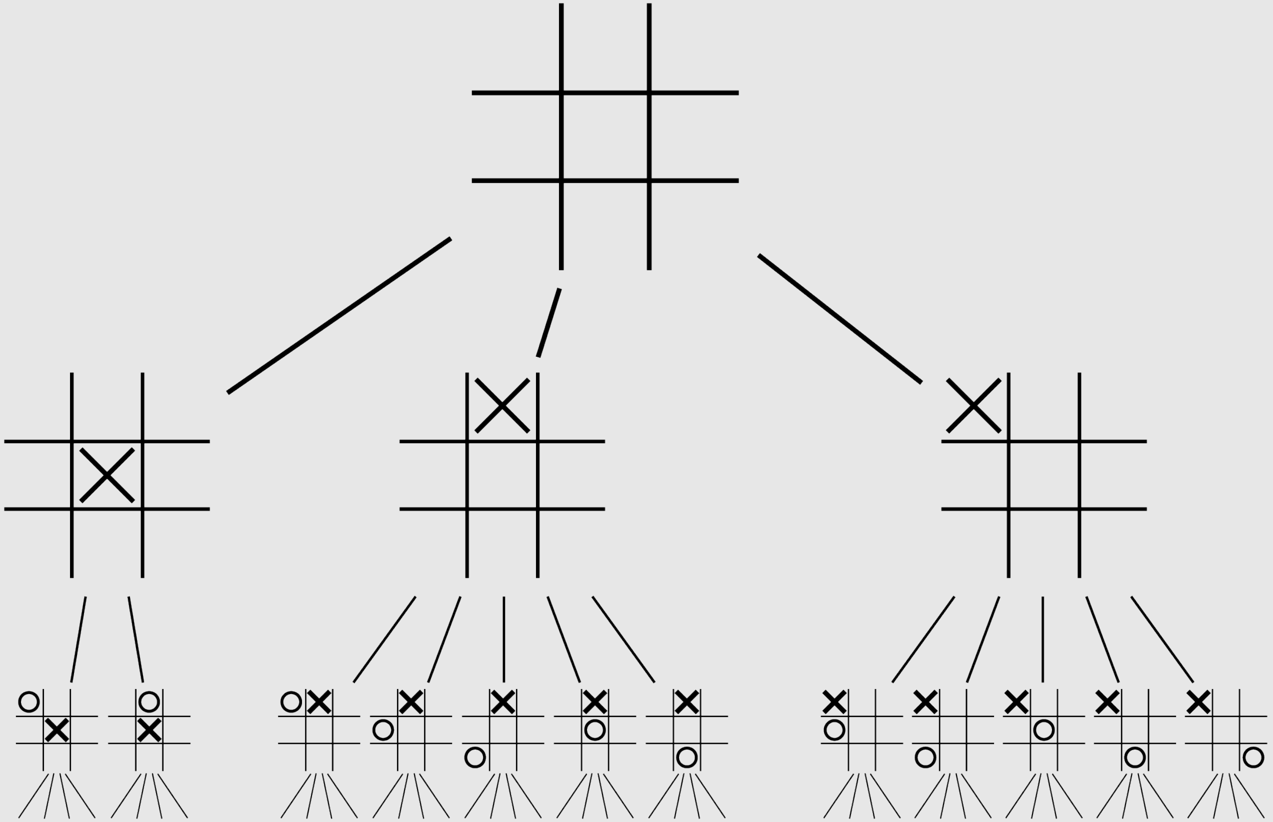
FIT5226
Multi-agent System & Collective Behaviour

Wk 9: Classical Game Theory (Extensive Form)

www.monash.edu.au

Bernd Meyer, May 2025

Extensive Form - The Idea



Extensive Form

- The normal form game representation does not incorporate any notion of sequence, or time, of the actions of the players

- The extensive form is an alternative representation that makes the **temporal structure** explicit (turn-taking)

- Two variants:

  - **perfect information** extensive-form game
    - *every player observes the **complete history** of the game*

  - **imperfect-information** extensive-form game

Extensive Form - Perfect Information

A (finite) perfect-information game (in extensive form) is defined by the tuple $(N, A, H, Z, \chi, \rho, \sigma, u)$, where:

- Players: N
- Actions: A
- Choice nodes: H
- Action function: $\chi : H \rightarrow 2^A$
- Player function: $\rho : H \rightarrow N$
- Terminal nodes: Z
- Successor function: $\sigma : H \times A \rightarrow H \cup Z$
- Utility function: $u = (u_1, \ldots, u_n)$; $u_i : Z \rightarrow R$ is a utility function for player i on the terminal nodes Z

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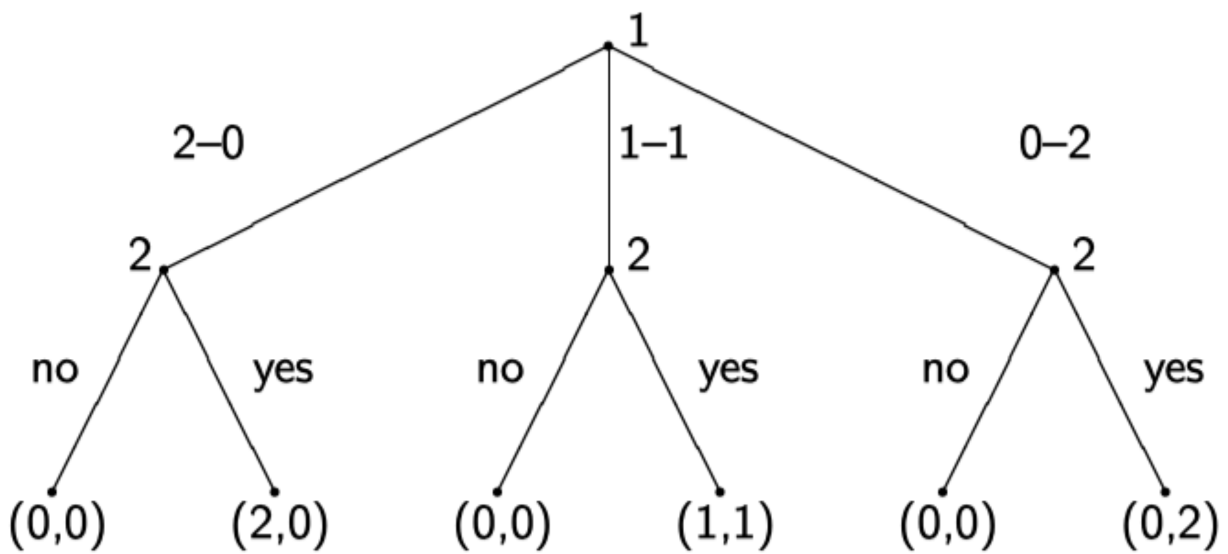
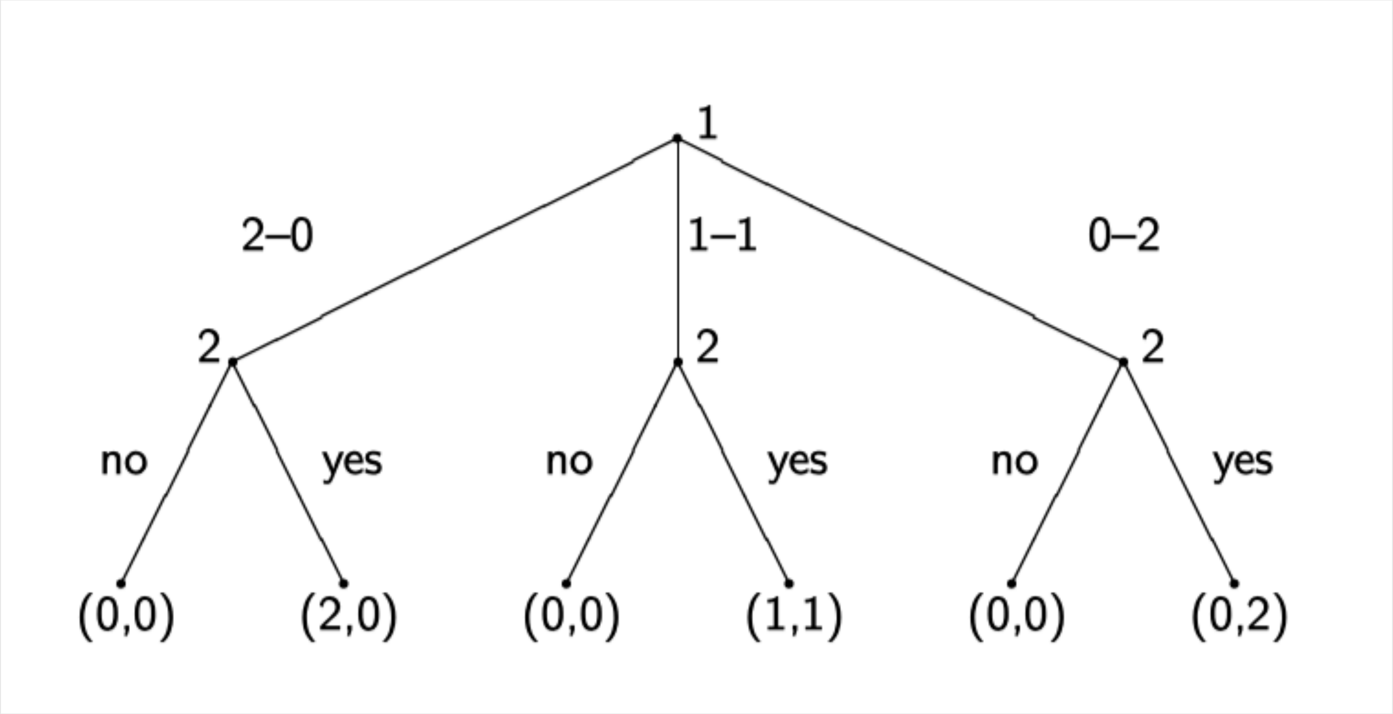
- Utility function: $u = (u_1, \ldots, u_n)$; $u_i : Z \to \mathrm{R}$ - payoff **only** in end states

Example: Sharing Game



Imagine a brother and sister following the following protocol for sharing two indivisible and identical presents from their parents. First the brother suggests a split, which can be one of three—he keeps both, she keeps both, or they each keep one. Then the sister chooses whether to accept or reject the split. If she accepts they each get their allocated present(s), and otherwise neither gets any gift.

Example: Sharing Game



How many pure strategies does each player have?

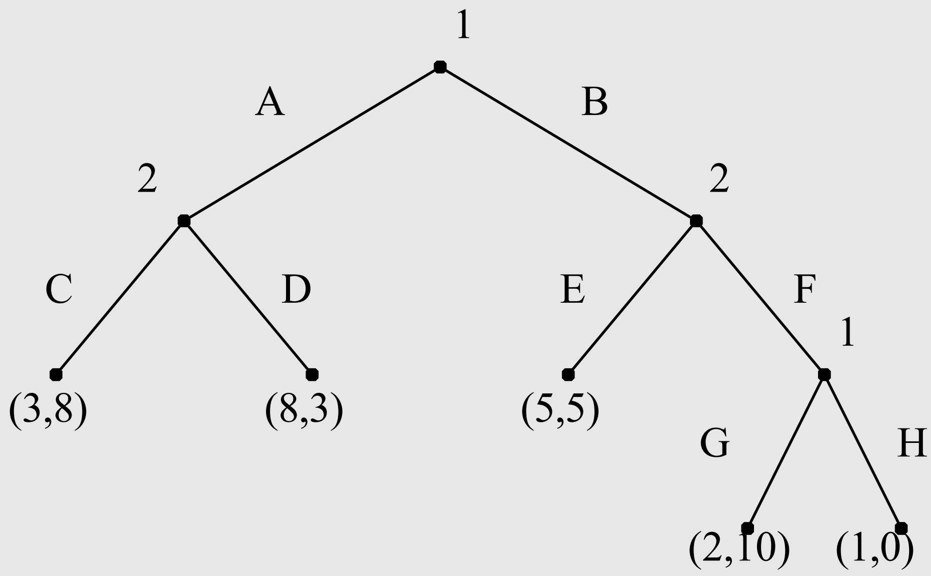
Pure Strategies

A pure strategy for a player in a perfect-information game is a complete specification of which deterministic action to take at every node where the player is active.

Definition (pure strategies)

Let $G = (N, A, H, Z, \chi, \rho, \sigma, u)$ be a perfect-information game in extensive-form. The *pure* strategies of player i consist of the cross product $\underset{h \in H, \rho(h)=i}{\times} \chi(h)$

Example

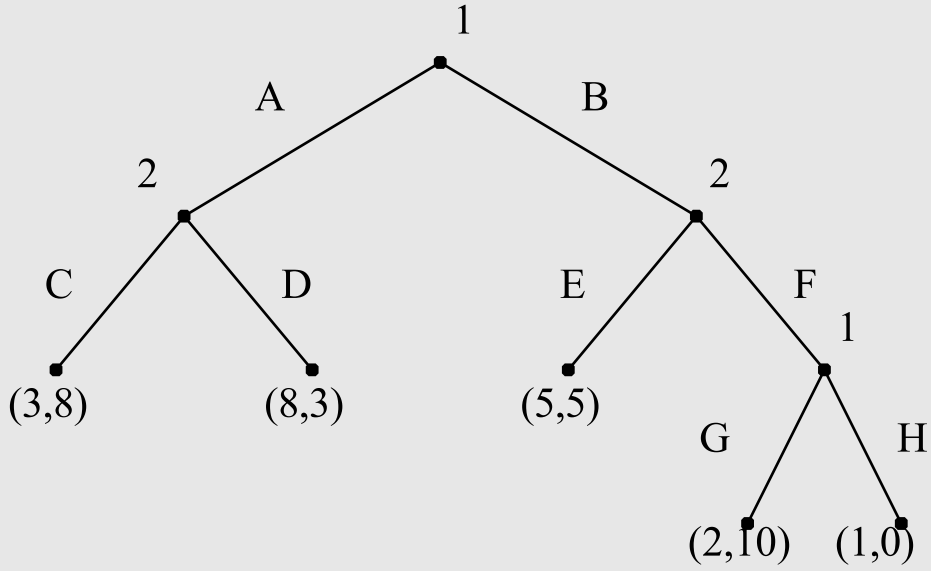


Pure Strategies for Player 1:

$$S_1 = \{(B,G);(B,H),(A,G),(A,H)\}$$

This is despite G, H never being played if A is chosen

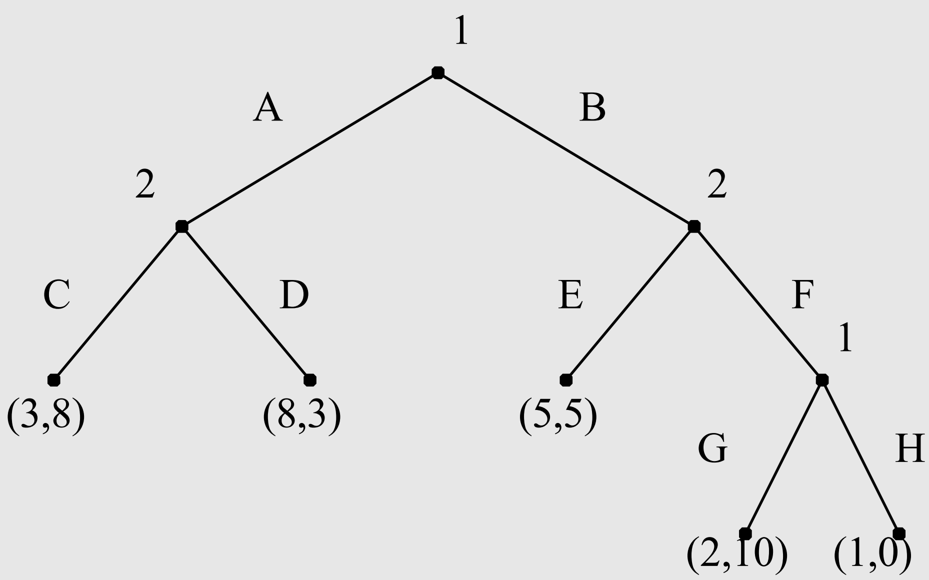
Example



Pure Strategies for Player 2:

$$S_2 = \{(C,E);(C,F);(D,E);(D,F)\}$$

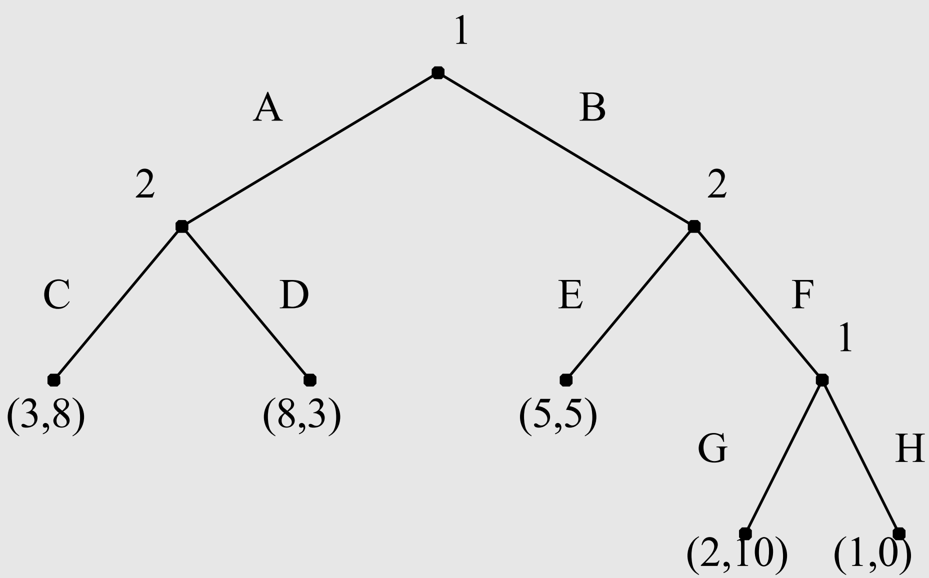
Induced Normal Form



	CE	CF	DE	DF
AG	3, 8	3, 8	8, 3	8, 3
AH	3, 8	3, 8	8, 3	8, 3
BG	5, 5	2, 10	5, 5	2, 10
BH	5, 5	1, 0	5, 5	1, 0

Now we can convert a perfect-information extensive form game back to normal form and reuse all our tools, incl. Nash Equilibrium

Pure Strategy Nash Equilibrium (PSNE) Example



	CE	CF	DE	DF
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Pure Strategy Equilibria?

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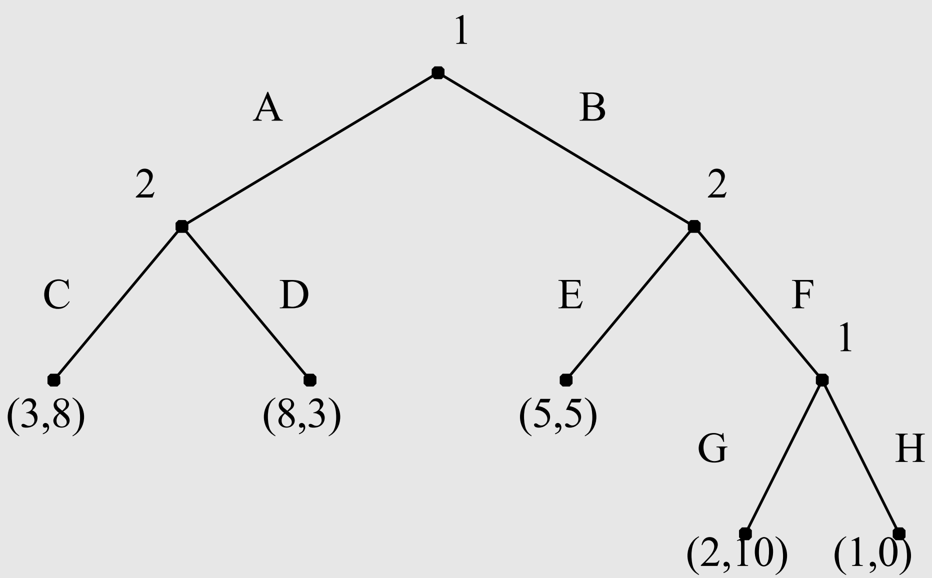

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Pure Strategy Equilibria?

AG / *CF*
AH / *CF*
BH / *CE*

PSNE for Extensive Form Games

Theorem: Every Perfect-Information Extensive Form Games has a Nash Equilibrium in **pure** strategies

Proof Idea: must be the case as moves are sequential and information is perfect, ie. at every move every player knows the complete history; the decision can be built step by step and no randomisation is required (or would even help).

Extensive Form vs Normal Form

- Advantage of extensive from: much more compact
- While every PIEF game has an induced NF game, the inverse is not true.

Can you prove this??

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Can you prove this??

For example, try to rewrite matching pennies in extensive form